

MOHITH DHARSHAN J

B.E. COMPUTER SCIENCE AND ENGINEERING

Chennai, Tamil Nadu, India

7845122655

jmohith2912@gmail.com

[GitHub: github.com/Mohith2912](https://github.com/Mohith2912)

[LinkedIn: linkedin.com/in/mohith-dharshan-994357381](https://www.linkedin.com/in/mohith-dharshan-994357381)

PROFILE

Sophomore B.E. Computer Science and Engineering student with practical experience in national-level and college-level hackathons and software projects. Experienced in frontend and backend development, Firebase, AI/ML integration, computer vision, VLM-based reasoning, and rapid product prototyping. Built solutions across healthcare, logistics, e-commerce, and industrial quality inspection. Strong interest in software engineering, Java programming, data structures, algorithms, and real-world AI applications.

HACKATHON & DEVELOPMENT EXPERIENCE

Frontend Developer — UniHealth, CMR Hackfest

3.0

Jan 2026

- Worked on a healthcare management solution covering hospital and pharmacy workflows.
- Built responsive interfaces using React and integrated Firebase for real-time application data.
- Implemented authentication and real-time updates while coordinating with the backend team for API integration.
- Contributed to debugging, testing, usability improvements, and hackathon presentation.

Backend Developer & AI Trainer — Codeathon

Hackathon

Feb 2026

- Developed backend services and REST APIs using Node.js for structured data exchange and workflow management.
- Integrated Firebase Realtime Database for synchronized application data and secure workflows.
- Contributed to AI training using a Rain Forest Algorithm classification workflow.
- Worked with the team to test, debug, and optimize backend and AI-related components.

SELECTED PROJECTS

ExtendQuality — AI-Powered Industrial Visual

Inspection

AI / CV / VLM

- Designed an industrial inspection workflow beginning with an Industrial Camera and OpenCV image enhancement.
- Uses YOLOv11 for automated defect detection and a Quality Intelligence Engine for confidence-based evaluation.
- Routes high-confidence results to automatic decisions while uncertain cases are passed to VLM reasoning.
- Uses VLM reasoning to provide contextual interpretation and support recommendation generation.
- Includes a Recommendation Engine and Inspection History Database for traceability and continuous quality analysis.
- KPI monitoring covers inspection performance, defect trends, confidence, and quality intelligence.

UniHealth — Hospital & Pharmacy Management

System

React / Firebase

- Developed healthcare interfaces for hospital and pharmacy availability, dashboards, authentication, and real-time data.
- Worked across frontend workflows and coordinated with backend modules for API integration and reliable data handling.

FairDispatch — Intelligent Dispatch Optimization

Node.js / AI

- Developed backend/API workflows for dispatch data and integrated Firebase for real-time synchronization.
- Contributed to AI/ML-based workload classification and data-driven task allocation for fairer dispatch decisions.

E-Commerce & Utility Web Applications

Web / Firebase / APIs

- Built responsive product interfaces, dynamic user workflows, search/filter functionality, and practical utility features.
- Integrated cloud services and APIs to support interactive, data-driven applications.

EDUCATION

SRM Easwari Engineering College — B.E.

Computer Science and Engineering

2025 – 2029

Sophomore | Semester 1: 8.53 / 10 | Semester 2: 9.16 / 10

- Relevant focus: Data Structures, Algorithms, Object-Oriented Programming, Software Development, AI/ML, and Web Technologies.

Sri Ramakrishna Matriculation Higher Secondary

School (SRM)

School

Higher Secondary Education

MOHITH DHARSHAN J

TECHNICAL RESUME — SOFTWARE / AI / FULL-STACK DEVELOPMENT

SKILLS

Programming Language	Java
Frontend	HTML, CSS, JavaScript, React, Responsive UI
Backend	Node.js, Express.js, REST APIs, API Integration
Database / Cloud	Firebase Realtime Database, Firebase Authentication
AI / ML	AI/ML Integration, Classification, Rain Forest Algorithm, VLM Workflows
Computer Vision	OpenCV, YOLOv11, Image Enhancement, Defect Detection
Development Tools	Git, GitHub, VS Code, npm, Figma
Core CS	Data Structures, Algorithms, OOP, Debugging, Problem Solving

EXTENDQUALITY — DETAILED PROJECT PROFILE

Industrial AI Quality Intelligence System

Objective: Automate industrial visual inspection and support quality decisions using computer vision and multimodal reasoning.

Architecture: Industrial Camera → OpenCV (Image Enhancement) → YOLOv11 (Defect Detection) → Quality Intelligence Engine → High-Confidence Auto Decision / VLM Reasoning → Recommendation Engine → Inspection History DB.

Key capabilities: defect detection, confidence-aware routing, VLM reasoning for ambiguous cases, actionable recommendations, historical inspection tracking, and KPI monitoring.

- Designed the system around a hybrid AI approach so deterministic/high-confidence detections can be handled automatically.
- Uses VLM reasoning for cases where visual context requires deeper interpretation instead of relying only on object-detection output.
- Inspection history enables analysis of recurring defects and quality trends.
- KPI monitoring is intended to provide visibility into defect rates, confidence, inspection throughput, and decision quality.

HACKATHON STRENGTHS

- Rapid prototyping and problem decomposition under hackathon time constraints.
- Cross-functional contribution across frontend, backend, AI/ML, computer vision, database, and presentation layers.
- Experience converting real-world problem statements into demonstrable software products.
- Team collaboration, Git/GitHub workflow, debugging, testing, and iterative development.
- Ability to explain technical architecture and project value during demonstrations and presentations.

CORE COMPETENCIES

Software Development	Frontend Engineering	Backend/API Development
AI/ML Integration	Computer Vision	VLM Reasoning
Firebase & Real-Time Data	Responsive UI	Git/GitHub
Problem Solving	Team Collaboration	Technical Presentation

LANGUAGES

English: Fluent **Tamil:** Fluent

PROFESSIONAL LINKS

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Email: jmohith2912@gmail.com

Phone: [+91 7845122655](tel:+917845122655)

References available upon request.